Venkata Lakshmi Parimala Pasupuleti

AI Engineer & Forward Deployed Engineer · Python / LangGraph / AWS · LLM Agents & Workflow Automation · Full-Stack & Consulting Delivery

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SUMMARY

AI Engineer with 9 years of full-stack production ownership. Built AgentProbe -- an open-source LLM orchestration and evaluation platform with multi-provider routing (GPT, Claude, Gemini/Groq, Ollama), CI-integrated regression gate, hallucination detection, and agent tracing. Built LeaseI -- an 8-agent LangGraph system with RAG, prompt engineering, workflow automation, and HITL guardrails on regulated decisions, evaluated across 1,240 test cases at 93.6% precision. At Magizhs Tech Solution, designed and delivered a production LangGraph agentic AI system for an external client from architecture through deployment -- LLM selection, RAG layer, validation guardrails, and integration with enterprise data sources. Seven years shipping enterprise full-stack systems at Entrata (Python, JavaScript/TypeScript, React, Angular, PostgreSQL, RabbitMQ, AWS) at 100K+ MAU. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

SKILLS

AI & LLM Engineering: Python, LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, Large Language Models (LLMs), LLM Integration, AI-Enabled Solutions, Intelligent Agents, Agent-Based Systems, Workflow Automation, Generative AI, LLM-Powered Applications, Prompting & Orchestration (GPT, Claude, Gemini), Prompt Engineering, RAG Pipelines, Vector Databases, ChromaDB, Embeddings, HITL, Guardrails, Output Validation, Hallucination Detection, Agent Tracing, LLMOps, Fine-tuning (HuggingFace, PyTorch), NLP, BERT, spaCy

Full-Stack & Backend: Python, JavaScript, TypeScript, Java, SQL, React.js, Angular, FastAPI, Flask, REST APIs, Microservices, Distributed Systems, RabbitMQ, Redis, Docker, Kubernetes (K8s), CI/CD (GitHub Actions), AWS (S3, EC2, RDS, Glue, Lambda), Azure, Google Cloud Platform (GCP)

Data & Infrastructure: PostgreSQL, MySQL, Oracle, SQL Server, Apache Spark, ETL, Data Pipelines, scikit-learn, PyTorch, TensorFlow, XGBoost, Tableau, Power BI, HIPAA Compliance, PHI Data Handling, Libsodium, Git, Linux, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI Engineer (Client Delivery)

- Designed and delivered a production LangGraph agentic AI system for an external client -- drove the full technical roadmap from architecture through deployment, including LLM selection, RAG layer over regulatory corpus, validation guardrails, and integration with enterprise data sources.
- Built a multi-step workflow automation pipeline (parse → plan → retrieve → validate → conditional routing) that enforced compliance rules deterministically before any downstream export, grounding every decision in cited source regulation.
- Implemented prompt engineering strategy for structured output extraction and chain-of-thought compliance reasoning; engineered guardrails blocking non-compliant results before delivery to client systems.

Entrata (PropTech SaaS · 100K+ MAU · enterprise full-stack delivery)

Sep 2018 - Aug 2024

Senior Software Engineer

- Shipped 20+ full-stack production modules (Python, JavaScript/TypeScript, React, Angular, PostgreSQL, RabbitMQ) across 120+ Agile sprints supporting 100K+ monthly active users; led microservices redesign with Docker (50% scalability improvement, 30% deployment time reduction).
- Owned compliance-critical data systems (180,000 HUD records, housing eligibility for 100K+ residents) -- zero violations across 7 years; baseline validation and rollback on every release cycle.
- Led AWS S3 document migration with full PHI compliance -- Libsodium-encrypted distributed pipelines, zero security incidents; built Python/Bash automation eliminating 80% of manual operational workload.
- Optimized 200+ SQL queries (30% performance gain); mentored 5+ engineers; conducted 300+ peer code reviews; built internal debugging tooling that reduced investigation time by 40%.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Fine-tuned a custom classification model architecture using HuggingFace Transformers and PyTorch; deployed inference API on Quart; presented model design, training methodology, and evaluation results to faculty.
- Built multi-source ML prediction models (Python, R, $R^2 = 0.977$); applied Isolation Forest anomaly detection across 3 research pipelines; delivered Python + database workshops to 600+ students.

PROJECTS

AgentProbe: LLM Orchestration & Evaluation Platform

2025 | github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Multi-provider routing: GPT, Claude, Gemini/Groq, Ollama

- Open-source LLM orchestration and evaluation platform: multi-provider routing (GPT, Claude, Gemini/Groq, Ollama), CI-integrated regression gate, LLM-as-judge scoring, hallucination detection, agent tracing, and pass@k harness. 5-metric eval engine; 15-endpoint FastAPI backend; evaluated LeaseI across 1,240 production-quality test cases.

Leasel: 8-Agent Workflow Automation System with RAG & HITL

2026 | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python · LangGraph (8-agent) · FastAPI · RAG · OCR · ChromaDB · SQLite · Streamlit

- Production agentic AI system: 8 LangGraph agents with RAG retrieval, prompt engineering, structured data extraction, workflow automation, and HITL guardrails on regulated decisions. 93.6% precision, 87.9% F1 across 1,240 test cases -- LLM reasons, deterministic/human logic makes every final regulated call.

AI on FHIR: HIPAA-Compliant Healthcare AI System

2024 | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask · React · LLMs · FHIR APIs · spaCy · BERT · Docker · HIPAA · RBAC

- HIPAA-compliant AI system over structured PHI data -- RBAC, LLM + BERT + embeddings integration, NLP-to-FHIR API translation (EN/ES); reduced clinical retrieval time ~60%.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — Graduated May 2026 | GPA: 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017